Cambridge Assessment International Education

Cambridge International AS & A Level

COMPUTER SCIENCE **9618/12**

Paper 12 Theory Fundamentals **March 2025**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **9** printed pages.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)	**1 mark** for 1 correct line **2 marks** for 2 or 3 correct lines **3 marks** for all 4 correct lines **Term** / **Definition** Pixel — The smallest identifiable component of an image Bit depth — The number of bits used to represent each colour Image resolution — The number of pixels wide by the number of pixels high File header — Stores data about the image file, e.g. file format, number of bits per pixel, file size	**3**
1(b)(i)	8	**1**
1(b)(ii)	**1 mark** for working • **10** * **5** * **8** (bits) / **8** // = **50** (pixels) * **8** (bits) / **8** **1 mark** for answer • 50 (bytes)	**2**
1(c)	**1 mark** per point • Increasing the colour depth results in increased <u>file</u> size // Decreasing the colour depth results in smaller <u>file</u> size • Increasing the colour depth means more **bits per pixel** and hence more data stored // Decreasing the colour depth means fewer **bits per pixel** and hence less data stored	**2**
1(d)	**1 mark** per point • Use run-length encoding // RLE • Record the colour <u>Blue</u>, and the number of times it occurs <u>10</u>	**2**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>2(a)</td><td>1 mark per point to max 2

• The lane detection system is built into / integrated into the car

• The lane detection system only performs one task
• The lane detection system is not easily changed/updated by the car owner</td><td>2</td></tr>
<tr><td>2(b)</td><td>1 mark for primary
• e.g. Miles travelled in the current journey, before the engine is turned off

1 mark for secondary
• e.g. Total miles travelled since the car was built // miles for most recent journey after engine switched off</td><td>2</td></tr>
<tr><td>2(c)</td><td>1 mark for all correct ticks

<table>
<tr><th>Statement</th><th>True</th><th>False</th></tr>
<tr><td>The screen always has five different layers</td><td></td><td>✓</td></tr>
<tr><td>A processor determines the horizontal and vertical coordinates of the point of contact</td><td>✓</td><td></td></tr>
<tr><td>The touchscreen will work if any object touches the screen</td><td>✓</td><td></td></tr>
</table></td><td>1</td></tr>
</table>

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>3(a)</td><td>1 mark for correct opcode and 1 mark for corresponding operand

OR #255 // OR #154 // XOR #154

e.g.
• OR...
• ... #255</td><td>2</td></tr>
<tr><td>3(b)</td><td>1 mark for correct opcode and 1 mark for corresponding operand

XOR #255

e.g.
• XOR...
• ... #255</td><td>2</td></tr>
<tr><td>3(c)</td><td>7E</td><td>1</td></tr>
<tr><td>3(d)</td><td>11110000</td><td>1</td></tr>
</table>

Question	Answer	Marks
3(e)	**1 mark** per point • Correct conversion to binary 01111111 (127) and 00001100 (12) • Working e.g. turning 01111111 into two's complement 10000001 • Answer: 1000 1101	**3**

Question	Answer	Marks
4(a)	**1 mark** per point to **max 2** e.g. • Attempts to **translate** the whole source code • Creates a separate error report at the end of the translation process • If translation successful / no errors creates an **executable** file	**2**
4(b)	**1 mark** per point to **max 2** e.g. • Reads each line then translates it **and executes** it • Stops when an error is encountered // displays errors where it finds them	**2**
4(c)	**1 mark** per point, **max 2** for writing, **max 2** for testing Writing e.g. • Enter code into an editor • Pretty printing to identify key terms • Context-sensitive prompts to help complete statements • Expand and collapse code blocks • Auto-complete to suggest what to type next • Auto-formatting to indent code blocks • Dynamic syntax checking Testing e.g. • Single stepping to run the code line by line • Breakpoints to stop the code at set points to check values • Report window to see how variables change	**4**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>5</td><td>1 mark for first 4 rows, 1 mark for second 4 rows (shaded)

<table>
<tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td></td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr>
</table></td><td>2</td></tr>
</table>

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>6</td><td>1 mark for identification of an application
Max 2 marks for relevant description
e.g.
• Police identifying wanted people
• Uses image recognition
• ... to identify features/characteristics/items in an image

• Natural language interfaces
• Use speech recognition to identify words that are spoken
• ... and adapts to learn regional accents

• Self-driving cars
• Detects its position on the road and within the traffic
• Follows a route // Collision avoidance // Self-parking etc.

• Spoken Interfaces
• Use natural language processing
• ... to take a sentence and work out its meaning

• Game playing
• Models characters in a computer game
• ... to allow computer characters to react according to the player's movements</td><td>6</td></tr>
</table>

Question	Answer	Marks
7(a)(i)	**1 mark** for security measure **1 mark** each to **max 2** for how the chosen measure works: • Firewall • Checks **incoming** connections • ... against criteria • Blocks data from entering specific ports • Blocks data that does not meet whitelist that meets blacklist • Proxy server • Prevents devices accessing the web server directly • Intercepts any requests • Forwards the request using its own IP address • Screens returning data before sending it to the user	**3**
7(a)(ii)	**1 mark** for security measure **1 mark** each to **max 2** for description of the chosen measure: • Encryption • Encodes/scrambles data • ... so if it is intercepted it cannot be **understood** • Algorithm/key is required to decode the data	**3**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>7(b)</td><td>1 mark for characteristic
1 mark for description of application to examination software:

<table>
<tr><th>Thin-client characteristic</th><th>Description of use in this software</th></tr>
<tr><td>Data is not stored on the client computer</td><td>Exam papers are stored on the server and not on the examiner’s computer // exam papers are not permanently stored on the examiners’ computers</td></tr>
<tr><td>Client computer is reliant on access to server</td><td>Examiners cannot mark if their device cannot access the server / the server ‘goes down’</td></tr>
<tr><td>Client computer heavily reliant on network/internet connection</td><td>The marking software will not operate without network/internet access</td></tr>
<tr><td>Client computer requires few local resources/memory</td><td>Examiners can use devices with low resources and the marking software will still function</td></tr>
<tr><td>Client computer performs minimal functions/processes</td><td>The marking software transmits requests, the server responds and sends the response to the user</td></tr>
</table></td><td>4</td></tr>
<tr><td>7(c)(i)</td><td>1 mark each to max 2:

• Receives packets from internet
• Analyses the destination IP address of each packet
• Forwards the packet towards its destination
• ... using the routing table
• Maintains/updates the routing table
• Finds the most efficient route to the destination</td><td>2</td></tr>
<tr><td>7(c)(ii)</td><td>1 mark each to max 2:

• The PSTN consists of many different types of communication lines
• … therefore the digital data may need to be converted into a different form/analogue signal
• Data is transmitted in both directions at the same time // duplex data transmission
• Using a PSTN the communication passes through different switching centres/ISPs</td><td>2</td></tr>
</table>

Question	Answer	Marks
8(a)	**1 mark** for: 1-to-many	**1**

8(b)	**1 mark** each:
	
	• Creating table EXAM with opening and closing brackets
	• All fields with appropriate data types and commas at end of lines
	• ExamID as primary key
	
	Example:
	`CREATE TABLE EXAM(`
	`ExamID varchar NOT NULL,`
	`Subject varchar,`
	`Level int,`
	`TotalMarks int,`
	`PRIMARY KEY(ExamID)`
	`);`
		3

8(c)	**1 mark** each:
	
	• Altering table EXAM_QUESTION
	• Linking ExamID to ExamID in EXAM
	
	Example.
	`ALTER TABLE EXAM_QUESTION`
	`ADD FOREIGN KEY (ExamID) REFERENCES EXAM(ExamID);`
		2

8(d)	**1 mark** each to **max 5**: • STUDENT table identified with suitable Primary Key • A linking table between STUDENT and EXAM with suitable Primary Key **and** appropriate name • … that includes the Primary Key of the STUDENT table as a Foreign Key to join with STUDENT • ... and includes the Primary Key of the EXAM table as a Foreign Key to join with EXAM • A linking table between STUDENT and EXAM_QUESTION with suitable Primary Key **and** appropriate name • … that includes the Primary Key of Table 2 as a Foreign Key to join with Table 2 • … that stores the ExamQuestionID and the mark for that question	**5**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>9(a)</td><td>1 mark each to max 2:

• Creates an executable file
• … so the code can be tested multiple times without having to recompile
• … so repeated testing takes less time</td><td>2</td></tr>
<tr><td>9(b)</td><td>1 mark for identification of each feature and 1 mark for matching description:

e.g.
For coding:
<table>
<tr><th>IDE feature</th><th>Description</th></tr>
<tr><td>Context-sensitive prompts</td><td>Gives suggestions for code as the user types instead of having to write/remember the code</td></tr>
<tr><td>Auto-correct</td><td>Corrects spelling mistakes so that user has fewer errors to correct</td></tr>
</table>
For presentation:
<table>
<tr><th>IDE feature</th><th>Description</th></tr>
<tr><td>Pretty-printing</td><td>Colour code keywords so the user can identify any errors</td></tr>
<tr><td>Expand/collapse (code) blocks</td><td>The user can hide code that they are not currently working on</td></tr>
</table>
For debugging:
<table>
<tr><th>IDE feature</th><th>Description</th></tr>
<tr><td>Single stepping</td><td>Run the code one line at a time // shows the effect of each line of code</td></tr>
<tr><td>Breakpoints</td><td>Stop the code running at a set point to check the flow/variable contents</td></tr>
</table></td><td>6</td></tr>
<tr><td>9(c)</td><td>1 mark each to max 2:

• Saves programming/testing time as code does not have to be written/re-written from scratch // code does not have to be tested
• Code is already tested so it is more robust/likely to work
• The programmer does not need to maintain the library // library routines are updated automatically
• Can perform complex calculations that the programmer may be unable to do
• Makes code more easily readable</td><td>2</td></tr>
</table>